

PADHAISHURU

GATE Preparation Platform

GATE EC

Electronics & Communication Engineering

TREND-BASED MOCK PAPER

— 01 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	9 subjects

Difficulty Distribution

Easy: 27 questions (42%)
Moderate: 30 questions (46%)
Difficult: 8 questions (12%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE EC — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

2M [-0.66]

[General Aptitude]

Sum of first 100 natural numbers:

- (A) 1000
- (B) 5050
- (C) 4950
- (D) 5000

Q2.

2M [no -ve]

[General Aptitude]

The 12 musical notes are given as $\sharp, \flat, \sharp, \flat, \sharp, \flat, \sharp, \flat$, and $\flat$. Frequency of each note is 122 times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $\sharp$ and $\flat$ is:

*Enter your answer here***Q3.**

2M [-0.66]

[General Aptitude]

If train 100m crosses pole in 10s, speed:

- (A) 10 m/s
- (B) 36 km/h
- (C) 18 km/h
- (D) 72 km/h

Q4.

2M [-0.66]

*[General Aptitude]*For a real number $x > 1$, $\frac{1}{x} \frac{1}{x} \frac{1}{x} \log 2 + \log 3 + \log 4 = 1$ The value of x is

- (A) 4
- (B) 12
- (C) 24
- (D) 36

Q5.

2M [no -ve]

[General Aptitude]

Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify the missing word in Group-II. Group-I: Abuse Insult Ridicule Group-II: _____ Praise Appreciate

*Enter your answer here***Q6.**

2M [-0.66]

[General Aptitude]

Choose synonym of 'trivial':

- (A) Significant
- (B) Insignificant
- (C) Important
- (D) Major

Q7.

3M [-0.66]

[General Aptitude]

The greatest prime factor of (3199 - 3196) is

- (A) 13
- (B) 17
- (C) 3
- (D) 11

Q8.

2M [-0.66]

[General Aptitude]

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration' is: Note: The figures shown are representative. 1 Iteration 0: Iteration 1: 1
1 Length of each segment: 3 3 1 1 Iteration 2: 9 Length of each segment: 9

- (A) A
- (B) B
- (C) C
- (D) D

Q9.

3M [-0.66]

[General Aptitude]

Which one of the following plots represents $() = - ||$, where is a non- zero real number? Note: The figures shown are representative.

- (A) A
- (B) B
- (C) C
- (D) D

Q10.

2M [-0.66]

[General Aptitude]

If 'ˆ' denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored _____ weary]. Which one of the given options is appropriate to fill the blank?

- (A) jaded
- (B) baffled
- (C) dead
- (D) worsted

Q11.

2M [no -ve]

[General Aptitude]

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above sentence.

Enter your answer here

Q12.

2M [no -ve]

[General Aptitude]

Choose opposite of 'benevolent':

Enter your answer here

Q13.

2M [no -ve]

[Engineering Mathematics] $i^4 =$:

*Enter your answer here***Q14.**

1M [no -ve]

[Engineering Mathematics] $P(A \cap B) = ?$

*Enter your answer here***Q15.**

1M [no -ve]

*[Engineering Mathematics]*Integral of $1/x$ dx:

*Enter your answer here***Q16.**

2M [-0.66]

[Network, Signals & Systems]

A full adder and an XOR gate are used to design a digital circuit with inputs X, Y, and Z, and output F, as shown below. The input Z is connected to the carry-in input of the full adder. If the input Z is set to logic '1', then the circuit functions as _____ with X and Y as inputs.

- (A) an adder
- (B) a subtractor
- (C) a multiplier
- (D) a binary to Gray code converter

Q17.

2M [-0.66]

[Network, Signals & Systems]

Consider the unity-negative-feedback system shown in Figure (i) below, where gain $K = 0$. The root locus of this system is shown in Figure (ii) below. For what value(s) of K will the system in Figure (i) have a pole at $-1 + j$?

- (A) $K = 5$
- (B) $K = 1/5$
- (C) For no positive value of K
- (D) For all positive values of K

Q18.

2M [-0.66]

[Network, Signals & Systems]

The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the points A, B, and C mark three different positions. Which of the following statements is/are true?

- (A) For between B and C, the electron diffusion current is directed from C to B.
- (B) For between B and A, the electron drift current is directed from B to A.
- (C) For between B and C, the electric field is directed from B to C.
- (D) For between B and A, the electric field is directed from A to B.

Q19.

2M [no -ve]

[Network, Signals & Systems]

As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is then closed at $t = 0$. The total energy dissipated in the ideal Zener diode ($V_Z = 5$ V) after the switch is closed (in mJ, rounded off to three decimal places) is _____.

Electronics and Communication Engineering (EC) Q.36 – Q.65 Carry TWO marks Each

- (A) A
- (B) B
- (C) C
- (D) D

Q20.

2M [-0.66]

[Network, Signals & Systems]

A 50 lossless transmission line is terminated with a load Z_L of $(50 - j75) \Omega$. If the average incident power on the line is 10 mW, then the average power delivered to the load (in mW, rounded off to one decimal place) is _____.

- (A) A
- (B) B
- (C) C
- (D) D

Q21.

2M [-0.66]

[Network, Signals & Systems]

A stick of length one meter is broken at two locations at distances of 1 and 2 from the origin (0), as shown in the figure. Note that $0 < 1 < 2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces? Note: All lengths are in meter. The figure shown is representative.

- (A) $1 < 0.5$
- (B) $2 > 0.5$
- (C) $2 < 1 + 0.5$
- (D) $1 + 2 < 1$

Q22.

2M [no -ve]

[Network, Signals & Systems]

The diode in the circuit shown below is ideal. The input voltage (in Volts) is given by $v_i = 10 \sin 100t$, where time is in seconds. The time duration (in ms, rounded off to two decimal places) for which the diode is forward biased during one period of the input is _____.

Enter your answer here

Q23.

2M [-0.66]

[Network, Signals & Systems]

The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16} \text{ /m}^3$ at 300 K. If the electron and hole mobilities are $0.15 \text{ m}^2/\text{Vs}$ and $0.05 \text{ m}^2/\text{Vs}$, respectively, then the intrinsic resistivity of the semiconductor (in k.m) at 300 K is _____. (Charge of an electron = $1.6 \times 10^{-19} \text{ C}$.)

- (A) 1.65
- (B) 1.25
- (C) 0.85
- (D) 1.95

Q24.

1M [-0.33]

[Electronic Devices]

SCR has:

- (A) 2 terminals
- (B) 3
- (C) 4
- (D) 5

Q25.

1M [-0.33]

[Electronic Devices]

n-type majority carriers:

- (A) Holes
- (B) Electrons
- (C) Both equal
- (D) None

Q26.

1M [-0.33]

[Electronic Devices]

MOSFET is:

- (A) BJT
- (B) Unipolar
- (C) Bipolar
- (D) None

Q27.

1M [-0.33]

[Electronic Devices]

Zener diode used for:

- (A) Rectification
- (B) Voltage regulation
- (C) Amplification
- (D) None

Q28.

1M [-0.33]

[Analog Circuits]

Common emitter has:

- (A) High input Z
- (B) Low input Z
- (C) Zero gain
- (D) Unity

Q29.

1M [no -ve]

[Analog Circuits]

Ideal op-amp open-loop gain:

Enter your answer here

Q30.

1M [-0.33]

[Analog Circuits]

FET is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q31.

1M [-0.33]

[Analog Circuits]

;" —â \$¥B —3

- (A) ; òf ³±
- (B) ; òf Ó±
- (C) lc/lb
- (D) le/lc

Q32.

1M [-0.33]

[Analog Circuits]

RC oscillator frequency:

- (A) $f=1/(2 \times 2 \times \pi RC)$
- (B) $f=1/RC$
- (C) $f=1/(4 \times 2 \times \pi RC)$
- (D) $f=2 \times \pi RC$

Q33.

1M [no -ve]

[Digital Circuits]

ROM is:

*Enter your answer here***Q34.**

1M [-0.33]

[Digital Circuits]

D flip-flop follows input after:

- (A) Same instant
- (B) One clock
- (C) Two clocks
- (D) Never

Q35.

1M [-0.33]

[Digital Circuits]

4-bit binary counter counts:

- (A) 0-15
- (B) 0-9
- (C) 1-16
- (D) 1-10

Q36.

1M [-0.33]

[Digital Circuits]

TTL stands for:

- (A) Transistor-Transistor Logic
- (B) Tunnel Transistor Logic
- (C) Totem Pole Logic
- (D) None

Q37.

1M [-0.33]

[Control Systems]

Transfer function is defined:

- (A) In time domain
- (B) Laplace
- (C) Fourier
- (D) Z-domain

Q38.

1M [-0.33]

[Control Systems]

Nyquist criterion uses:

- (A) Polar plot
- (B) Root locus
- (C) Routh
- (D) Bode

Q39.

1M [-0.33]

[Control Systems]

Routh-Hurwitz determines:

- (A) Stability
- (B) Time response
- (C) Frequency
- (D) Sensitivity

Q40.

1M [-0.33]

[Control Systems]

Lead compensator:

- (A) Improves steady state
- (B) Improves transient
- (C) Both
- (D) None

Q41.

1M [no -ve]

[Communication Systems]

ASK modulation varies:

Enter your answer here

Q42.

1M [-0.33]

[Communication Systems]

FM has:

- (A) Constant amplitude
- (B) Variable
- (C) Both
- (D) None

Q43.

1M [no -ve]

[Communication Systems]

PCM stands for:

*Enter your answer here***Q44.**

1M [-0.33]

[Communication Systems]

AM stands for:

- (A) Amplitude Modulation
- (B) Angle Modulation
- (C) Analog Modulation
- (D) None

Q45.

1M [no -ve]

[Electromagnetics]

Maxwell has how many equations:

*Enter your answer here***Q46.**

1M [-0.33]

*[Electromagnetics]*Characteristic impedance $Z_0 =$:

- (A) $\sqrt{L/C}$
- (B) L/C
- (C) C/L
- (D) $\sqrt{LC}$

Q47.

1M [no -ve]

[Electromagnetics]

Skin depth increases with:

*Enter your answer here***Q48.**

1M [-0.33]

[Electromagnetics]

Waveguide supports:

- (A) TEM
- (B) TE/TM
- (C) Both
- (D) None

Q49.

1M [-0.33]

[Control Systems]

Transfer function is defined:

- (A) In time domain
- (B) Laplace
- (C) Fourier
- (D) Z-domain

Q50.

2M [no -ve]

[Network, Signals & Systems]

$s(s+1)(s+2)R(s) + E(s)G(s)Y(s) - R(s)$, $Y(s)$, and $E(s)$ are the Laplace transforms of $r(t)$, $y(t)$, and $e(t)$, respectively. If the input $r(t)$ is a unit step function, then _____.

Enter your answer here

Q51.

2M [no -ve]

[Network, Signals & Systems]

A 50 lossless transmission line is terminated with a load Z_L of $(50 - j75)$. If the average incident power on the line is 10 mW, then the average power delivered to the load (in mW, rounded off to one decimal place) is _____.

Enter your answer here

Q52.

2M [-0.66]

[Network, Signals & Systems]

In the circuit shown below, the transistors M_1 and M_2 are biased in saturation. Their small signal transconductances are g_{m1} and g_{m2} respectively. Neglect body effect, channel length modulation and intrinsic device capacitances. V_{DD} I_{ref} V_{DD} R_D M_1 R_B v_{out} R_S M_2 C_1 v_{in} Assuming that capacitor C_1 is a short circuit for AC analysis, the exact magnitude of small signal voltage gain $|A|$ is _____.

- (A) $g_{m2} R_D$
- (B) 1
- (C) 1
- (D) 1

Q53.

2M [-0.66]

[Network, Signals & Systems]

Let (x, y, z, t) and $u(x, y, z, t)$ represent density and velocity, respectively, at a point (x, y, z) and time t . Assume u is continuous. Let V be an arbitrary volume in space enclosed by the closed surface S and $\mathbf{n}$ be the outward unit normal of S . Which of the following equations is/are equivalent to $\frac{d}{dt} \int_V u \, dv = 0$?

- (A) $\frac{d}{dt} \int_V u \, dv = - \int_S u \mathbf{n} \cdot d\mathbf{s}$
- (B) $\frac{d}{dt} \int_V u \, dv = \int_S u \mathbf{n} \cdot d\mathbf{s}$
- (C) $\frac{d}{dt} \int_V u \, dv = - \int_S (u \mathbf{n}) \cdot d\mathbf{s}$
- (D) $\frac{d}{dt} \int_V u \, dv = \int_S (u \mathbf{n}) \cdot d\mathbf{s}$

Q54.

2M [-0.66]

[Network, Signals & Systems]

Consider a continuous-time finite-energy signal $x(t)$ whose Fourier transform vanishes outside the frequency interval $[-\omega_c, \omega_c]$, where ω_c is in rad/sec. The signal $x(t)$ is uniformly sampled to obtain $x[n] = x(nT)$. Here, $T = \frac{1}{f_s}$, $f_s = \frac{1}{T}$, with $\delta(t)$ being the Dirac impulse, $\omega_c > 0$, and $T > 0$. The sampled signal $x[n] \sin(n\omega_c T)$ is passed through an ideal lowpass filter $H(\omega)$ with cutoff frequency ω_c and passband gain 1. The output of the filter is given by _____.

- (A) $x(t)$ if $\omega_c < \frac{1}{2T}$
- (B) $x(t)$ if $\omega_c < \frac{1}{T}$
- (C) $x(t)$ if $\omega_c < \frac{2}{T}$

Q55.

1M [-0.33]

[Analog Circuits]

Common emitter has:

- (A) High input Z
- (B) Low input Z
- (C) Zero gain
- (D) Unity

Q56.

2M [no -ve]

[Network, Signals & Systems]

Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) this system? Choose the correct option(s).

- (A) A
- (B) B
- (C) C
- (D) D

Q57.

1M [-0.33]

[Communication Systems]

PCM stands for:

- (A) Pulse Code Modulation
- (B) Phase Code Modulation
- (C) Pulse Carrier Modulation
- (D) None

Q58.

2M [no -ve]

[Network, Signals & Systems]

A full scale sinusoidal signal is applied to a 10-bit ADC. The fundamental signal component in the ADC output has a normalized power of 1 W, and the total noise and distortion normalized power is 10 W. The effective number of bits (rounded off to the nearest integer) of the ADC is _____.

Enter your answer here

Q59.

2M [no -ve]

[Network, Signals & Systems]

Consider the unity-negative-feedback system shown in Figure (i) below, where gain 0. The root locus of this system is shown in Figure (ii) below. For what value(s) of will the system in Figure (i) have a pole at $-1 + j$?

Enter your answer here

Q60.

2M [no -ve]

[Network, Signals & Systems]

A satellite attitude control system, as shown below, has a plant with transfer function $G(s) = 1$ cascaded with a compensator $C(s) = K(s + \alpha)$, where α and $s^2 + 4$ are positive real constants. $R(s) \rightarrow C(s) \rightarrow G(s) \rightarrow Y(s)$
 - In order for the closed-loop system to have poles at $-1 \pm j3$, the value of K must be _____.

Enter your answer here

Q61.

1M [-0.33]

[Engineering Mathematics] $P(A \cap B) = ?$

- (A) $P(A) + P(B)$
- (B) $P(A) + P(B) - P(A \cap B)$
- (C) $P(A)P(B)$
- (D) $P(A)/P(B)$

Q62.

2M [no -ve]

[Network, Signals & Systems]

Which of the following can be used as an n-type dopant for silicon? Select the correct option(s).

- (A) Arsenic
- (B) Boron
- (C) Gallium
- (D) Phosphorous

Q63.

1M [-0.33]

[Network, Signals & Systems]

Sequence the following sentences (P, Q, R, S) in a coherent passage: P: Shifu's student exclaimed, "Why do you run since the bull is an illusion?" Q: Shifu said, "Surely my running away from the bull is also an illusion." R: Shifu once proclaimed that all life is illusion. S: One day, when a bull gave him chase, Shifu began running for his life.

- (A) SPRQ
- (B) SRPQ
- (C) RSPQ
- (D) RPQS

Q64.

2M [-0.66]

[Network, Signals & Systems]

The relationship between any N-length sequence $x[n]$ and its corresponding N-point discrete Fourier transform $X[k]$ is defined as $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$. Another sequence $y[n]$ is formed as below $y[n] = \{ \{ \{ x[n] \} \} \}$. For the sequence $x[n] = \{1, 2, 1, 3\}$, the value of $y[0]$ is _____. Electronics and Communication Engineering (EC)

- (A) A
- (B) B
- (C) C
- (D) D

Q65.

2M [no -ve]

[Network, Signals & Systems]

Which of the following statements is/are TRUE with respect to ideal MOSFET- based DC-coupled single-stage amplifiers having finite load resistors?

- (A) The common-gate amplifier has an infinite input resistance
- (B) The common-source amplifier has an infinite input resistance
- (C) The input and output voltages of the common-source amplifier are in phase
- (D) The input and output voltages of the common-drain amplifier are in phase